

# INHIBITORS OF BACTERIAL PROTEIN SYNTHESIS

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- **Macrolides**
- **Tetracyclines**
- **Aminoglycosides**
- **Clindamycin**
- **Oxazolidinones**

## Macrolides

**Erythromycin (prototype), azithromycin, clarithromycin, roxithromycin**

*Mechanism of action:*

- *They bind* **reversibly to the 50S ribosomal subunit** *and inhibit bacterial protein synthesis.*
- *They are* **bacteriostatic** *in low concentrations and bactericidal in high concentrations.*

*Antibacterial spectrum:*

*Macrolides are effective against a number of organisms, including* **gram-positive bacteria** *e.g. pneumoncocci, staphylococci, streptococcus species, C. diphtheria, some* **gram- negative bacteria** *e.g. Neisseria, H. influenza, and* **intracellular microorganisms** *(Mycoplasma species, legionella and Chlamydia).*

## Pharmacokinetics:

- *Absorption of erythromycin is affected by food and HCl.*
- *Newer macrolides are acid stable and can be given orally.*
- *They reach all body fluids including prostate, placenta ( not teratogenic) and milk but only* **small amounts can penetrate to CSF.**
- *Azithromycin and roxithromycin are* **concentrated in neutrophils, macrophages and lung tissue***, so they have long t1/2 (given once daily) and they are* **useful against intracellular organisms.**
- *Macrolides are concentrated in liver and excreted primarily in active form via bile with only low levels found in urine.*
- *Azithromycin and clarithromycin are converted into active metabolites.*

## Adverse effects

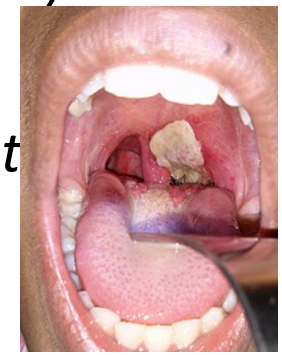
- *Gastrointestinal upset especially with erythromycin*
- *Cholestatic jaundice*
- *Ototoxicity*
- *QTc prolongation*

## Interactions:

- *Combination of the therapeutic dose of erythromycin with penicillin antagonizes the bactericidal effect of penicillin.*
- *Erythromycin decreases CYP450 enzymes, thus increase serum concentration of* **theophylline, oral anticoagulants, cyclosporins,** *and* **digoxin.**

## Therapeutic uses:

- *Treatment and prevention of infections caused by staphylococci, streptococci and pneumococci (gram-positive) in penicillin-hypersensitive individuals.*



- *Eradication of* **C. diphtheriae** *from pharyngeal carriers (1st choice).*
- *Azithromycin and roxithromycin are especially effective against intracellular and atypical bacteria (***chlamydia, mycoplasma, legionella***)*
- *Clarithromycin has activity against* **Toxoplasma***. It is also effective against* **Helicobacter pylori** *in peptic ulcer.*
- *It is a second line drug for the treatment of* **gonorrhea and syphilis.**
- *Treatment of middle ear and sinus infections, since the causative agents,* **H. influenza** *and* **Strep pneumonia** *are usually sensitive*

# TETRACYCLINES

## Oxytetracycline, doxycycline, minocycline, tigecycline

*Mechanism of action:*

- *Tetracyclines bind* **reversibly to the 30S ribosomal subunit** *and inhibit bacterial protein synthesis.*
- *They are* **bacteriostatic.**

*Antibacterial spectrum:*

*Tetracyclines display* **broad-spectrum activity** *and are effective against:*

*Most* **gram-positive**

*many* **gram-negative bacteria** *(Brucella species, Helicobacter pylori and Vibrio cholerae),*

**Intracellular organisms** *(Rickettsia, Mycoplasma and Chlamydia ).*

*Spirochetes (Borrelia burgdorfer), Actinomycines, Protozoa.*

## *Pharmacokinetics:*

- *Tetracyclines are adequately absorbed after oral ingestion*
- *Calcium (milk and Ca. antacids), magnesium (Mg hydroxide), aluminum hydroxide and iron interfere with their absorption.*
- *They reach all body fluids and placenta due to its lipid solubility but only small amounts can penetrate to CSF (Doxycycline is the most lipid-soluble).*
- *Because of their chelating properties with calcium they tend to be deposited in growing bones and teeth causing yellow discoloration of teeth.*
- *The primary route of elimination is renal. Doxycycline is excreted in stool and does not accumulate in patients with renal impairment.*

## *Therapeutic uses:*

- *Doxycycline is used as alternative to macrolides in Chlamydia, Mycoplasma, and Legionella infections.*
- *Borrelia burgdorferi: is the cause of* **Lyme disease**, *a spirochetal infection transmitted by infected ticks. The disease consists of skin rash, fever, and arthritis.*



- *Rickettsia: is a genus of small obligate intracellular bacteria causing* **typhus and Rocky Mountain spotted fever.**



- *Doxycycline and minocycline used for treatment of* **acne vulgaris.**
- *Doxycycline is effective in treatment of* **cholera**.



# *Adverse effects:*

- **Epigastric distress**
- **Yellow staining** *because of their chelating properties with calcium. This can occur after the fourth month of gestation or if they are given to children less than 8 years of age.*
- **Cholestatic hepatitis**
- **Phototoxicity:** *skin reaction in the form of sunburn.*
- **Superinfection** *with resistant staphylococci or C. difficile.*
- *Decreased synthesis of Vit. B12 leads to sore tongue & black hairy tongue*
- **Teratogenesis,** *when administered early in pregnancy.*



- *Contraindications:*
- *Severe hepatic disease and renal disease.*
- *It should be avoided in patients with peptic ulcer,*
- *In children (less than 8 years of age)*
- *In pregnancy and lactation.*

- *Interactions:*
- *Avoid simultaneous ingestion of dairy products (milk & cheese), antacids, or iron with tetracyclines because of chelating action of tetracyclines.*
- *Tetracyclines decrease vitamin K synthesis in intestinal lumen due to inhibition of intestinal flora leading to potentiation of oral anti-coagulant effect*

## *Tigecycline*

*a derivative of minocycline.*




- *Tigecycline exhibits* **broad-spectrum activity** *that includes methicillin-resistant staphylococci (MRSA), multidrug-resistant streptococci, vancomycin-resistant enterococci (VRE), B-lactamase producing gram-negative bacteria, Acinetobacter baumannii, and many anaerobic organisms.*
- *Tigecycline is not active against* **Proteus, Pseudomonas** *species.*

- *It is indicated for the treatment of complicated skin and soft tissue infections, complicated intra-abdominal infections, and community acquired pneumonia*
- *It penetrates tissues well but achieves low plasma concentrations. Consequently, tigecycline is a poor option for bloodstream infections.*
- *Tigecycline is associated with* **acute pancreatitis.**
- *tigecycline should be reserved for use in situations when alternative treatments are not suitable.*
- **Tigecycline may decrease the clearance of warfarin.**

# Aminoglycosides

**streptomycin, Gentamicin, tobramycin, amikacin, kanamycin, neomycin**

*Mechanism of action*

- *Aminoglycosides diffuse through porin channels in the outer membrane and an-oxygen dependent system transports the drug across the cytoplasmic membrane. only in* **Gram-negative aerobic spp***.*
- *Inside the cell, they* **irreversibly bind to 30S ribosomal subunit and inhibit bacterial protein synthesis.**
- *They are* **bactericidal.**
- *Bacterial killing is* **concentration-dependent**
- **A post-antibiotic effect** *is also present i.e., residual bactericidal activity is present after the serum concentration falls below the MIC.*

- *Antimicrobial spectrum:*
  *They have narrow spectrum against mainly Gram negative bacilli, very few Gram positive cocci, and Mycobacteria TB.*

- *Pharmacokinetics:*
- *All aminoglycosides are* **not absorbed orally** *(must be given parentrally) and* **cannot penetrate to CSF** *because they are highly polar compounds.*
- *They can cross the placental barrier and may cause* **congenital deafness.**
- *They are* **excreted unchanged** *in urine (dose adjustment is necessary in renal dysfunction). They become more active in alkaline urine.*

## *Therapeutic uses:*

**narrow spectrum** *and* **potential toxicity***.*

- *Streptomycin is currently used only for* **plague***,* **brucellosis** *and it is one of the 1st line drugs for* **TB***.*
- *Gentamicin and tobramycin are used in the treatment of* **gram-negative infections** *e.g. urinary tract and respiratory infections.*
- *They are used in* **gram-negative septicemia,** *usually combined with penicillin and/or metronidazole.*
- **Pseudomonas infections***: they are more efficient when combined with antipseudomonal penicillin*
- *A combination of* **vancomycin and gentamicin** *is useful in the treatment of* **enterococcal endocarditis***.*
- *Neomycin is used* **orally** *for* **hepatic encephalopathy** *to suppress intestinal bacteria that produce ammonia.*
- *Neomycin is used orally for* **sterilization of intestine** *before surgery*
- *Neomycin and gentamycin are used* **topically** *for skin and eye infections often in combination with polymixin B or bacitracin.*

*Adverse effects:*

*Aminoglycosides have a* **narrow therapeutic index**

- **Nephrotoxicity:**

  *It is increased by co-administration of other nephrotoxic drugs e.g.cephalosporins.*

- **Ototoxicity:**

  *It is increased by co-administration of other ototoxic drugs e.g. loop diuretics and in old age and with renal diseases.*

- **Neuromuscular block:.**

- **Allergic reactions:**

  *Contact dermatitis is a common reaction to topically applied neomycin.*

# Clindamycin

- *mechanism of action that is similar to that of the macrolides.*
- *Clindamycin is used primarily in the treatment of infections caused by* **gram-positive organisms, including MRSA and streptococcus, and anaerobic bacteria**
- *Clindamycin is available in both IV and oral formulations.*
- *It distributes well into all body fluids but* **exhibits poor entry into the CSF**.

- *Clindamycin is excreted into bile and urine.*
- *Low urinary excretion of active drug limits its clinical utility for urinary tract infections.*
- *Accumulation has been reported in patients with either severe renal impairment or hepatic failure.*
- *In addition to skin rash, the most common adverse effect is* **diarrhea***, which may represent a serious* **pseudomembranous colitis** *caused by overgrowth of C. difficile. Oral administration of either metronidazole or vancomycin is usually effective in the treatment of C. difficile infection.*

# Oxazolidinones

- **Linezolid and tedizolid**

## Mechanism of action

- *inhibiting the formation of the 70S initiation complex*

## Anti-bacterial spectrum

- *directed against gram-positive organisms, including* **methicillin-resistant Staphylococcus aureus, VRE, and penicillin-resistant streptococci.**
- *Corynebacterium species and Listeria monocytegen.*
- *It is also moderately active against Mycobacterium tuberculosis*
- *Linezolid is an alternative to daptomycin for infections caused by VRE.*
- **Because they are bacteriostatic, the oxazolidinones are not recommended as first-line treatment for MRSA bacteremia.**

## Pharmacokinetics

- *Linezolid and tedizolid are well absorbed after oral administration. IV formulations are also available.*
- *Linezolid is excreted mainly by renal route.*
- *Tedizolid its elimination occurs mainly via the liver.*

## Adverseeffects

- *nausea, diarrhea, headache, and rash.*
- *Linezolid and tedizolid possess* **nonselective monoamine oxidase activity** *and may lead to serotonin syndrome if given concomitantly with large quantities of tyramine-containing foods,*
- **Irreversible peripheral neuropathies and optic neuritis** *causing blindness have been associated with greater than 28 days of use.*

*Thank you*